ELASTIC CHICAGO MEETUP · MAY 13, 2026

Building Autonomous AI Agents

with Elasticsearch & LLMs

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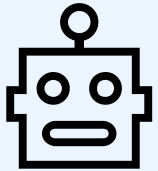
Elasticsearch | RAG / LLMs | Agent Builder



Elasticsearch
+
LLM Agents

Intro about myself

Deepesh Kumar

M.S. Artificial Intelligence @ Illinois Institute of Technology

- Graduate student studying AI, ML, and multi-agent systems at IIT Chicago
- Building LLM-powered autonomous agents with real-world retrieval and action capabilities
- Passionate about Elasticsearch, RAG pipelines, and production-grade AI systems
- Open to work — AI Solutions Engineer & Multi-Agent LLM Systems

Tonight's Talk

→ AI & Agents
→ Elasticsearch
→ RAG Pattern
→ Agent Builder
→ Live Demo

THE PROBLEM

It's 3am. Your service is down.

A human engineer checks 5 dashboards, greps through logs, correlates timestamps, maybe wakes their manager.

Siloed data

Logs, metrics, orders and events live in different systems. No single tool sees the full picture.

Slow investigation

Mean time to resolve incidents is measured in hours, not seconds — costing revenue and customers.

LLMs are blind

GPT-4 knows the world up to its training cutoff. It knows nothing about your data or last night's outage.

Solution: RAG + Agents

Combine Elasticsearch's retrieval with LLM reasoning. An agent that investigates automatically — in seconds.

CORE CONCEPTS

What is an AI Agent?

An agent perceives its environment, plans a course of action, executes tools, and loops until done.

Chatbot vs Agent

	Chatbot	Agent
Memory	None	**Long-term via ES**
Action	Text only	**Calls tools**
Planning	Single step	**Multi-step loop**
Data access	Training only	**Live retrieval**

The autonomous plan → act → observe loop

1. **Perceive**
   Receive user query or incident alert
2. **Plan**
   Decide which tools to call and in what order
3. **Act**
   Execute ES|QL queries, search, filter, fetch
4. **Observe**
   Read results — re-plan if more info needed

ELASTICSEARCH

Elasticsearch: The Memory Layer

Not just search — a distributed engine that gives agents instant recall across millions of documents.

SQL concepts map directly to Elasticsearch

SQL (what you know)	Elasticsearch	What it means
Database	**Cluster**	One or many machines working together
Table	**Index**	Optimised for search — auto-indexed fields
Row	**Document (JSON)**	Flexible schema — any JSON object
Column	**Field**	Every field searchable with no extra setup
Schema	**Mapping**	Define field types: text, keyword, vector
SELECT WHERE...	**Search query (JSON)**	BM25 + kNN + filters in one query

Key insight: Elasticsearch auto-indexes every field the moment a document is stored. Every field is instantly searchable — zero extra code.

RAG

Retrieval Augmented Generation

The open-book exam pattern — LLM reads the right document first, then answers. No hallucination.

1 User asks

"Why are payments failing?"

2 Embed query

Text → 384-dim vector

3 Hybrid search

BM25 + kNN + RRF in ES

4 Retrieve docs

Top-5 relevant chunks

5 LLM answers

Grounded, no hallucination

Why hybrid search?

Keyword (BM25)

Exact word match — great for IDs, error codes, product names

Vector (kNN)

Semantic similarity — finds meaning even when exact words differ

RRF Fusion

Blends both scores — consistently better results than either alone

BUILD OPTIONS

Two Ways to Build Agents

Choose based on your use case — code or no-code, both use Elasticsearch as the retrieval brain.

	Python + LangChain	Native Agent Builder
Setup time	Hours	**Minutes**
Code required	Full Python	**Zero code**
Infrastructure	You manage all	**Fully managed**
LLM models	Any via API key	**GPT-4o, Claude, Gemini**
Customisation	Unlimited	**ES|QL tools + skills**
Best for	Production apps	**Demos, POCs, enterprise**

Tonight's demo → Native Agent Builder inside Kibana. Zero code. Maximum impact on stage.

AGENT BUILDER

What's Inside Elastic Agent Builder

A complete platform for building, distributing and running AI agents — all inside Kibana.

GA

Tools

ES|QL queries agents can call. Built-in: search, esql_query, get_document. Custom: define your own.

GA

Skills

Instructions + tools for a domain. Load on demand — saves 21–39% of input tokens vs always-on prompts.

NEW

Plugins

Bundles of related skills distributed as a package. Install from GitHub or ZIP file.

GA

Agents

LLM + system prompt + skills + tools. One agent per use case. Share with your team or org.

PREVIEW

Workflows

Scripted automation steps + AI agent steps in one pipeline. Bidirectional — agent calls workflow and vice versa.

GA

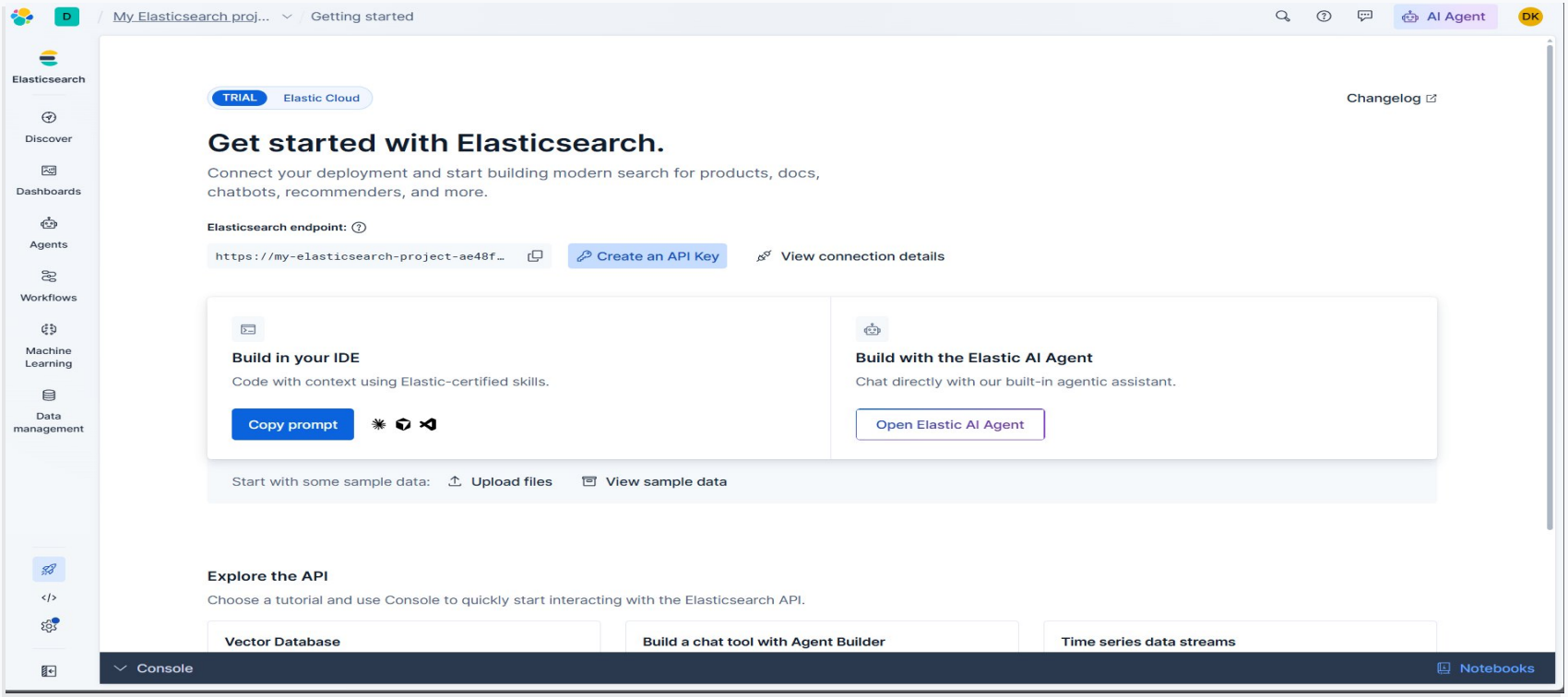
MCP & A2A

Connect Claude Desktop, Cursor, or LangChain to your Elastic agents via Model Context Protocol.

STEP 1 — SETUP

Elastic Cloud Serverless — Project Home

Starting point: Elasticsearch project with Agents, Workflows and Console all accessible from the sidebar.



Agents — your Agent Builder home

Workflows — automation engine

Console — run ES|QL queries

STEP 2 — DATA

Indexing Data via Kibana Console

Two Elasticsearch indices created: store-orders (13 records) and return-policies (6 documents). 200-OK confirms success.

My Elasticsearch proj... Developer Tools

Find apps, content, and more.

AI Agent

Console Search Profiler Grok Debugger

Shell History Config

Click to send request

Export requests Import requests

```
PUT /store-orders
{
  "mappings": {
    "properties": {
      "order_id":        { "type": "keyword" },
      "customer_name":   { "type": "keyword" },
      "customer_email":  { "type": "keyword" },
      "product_name":    { "type": "text" },
      "product_category":{ "type": "keyword" },
      "amount":          { "type": "float" },
      "status":          { "type": "keyword" },
      "order_date":      { "type": "date" },
      "delivery_date":   { "type": "date" },
      "return_requested":{ "type": "boolean" },
      "return_count":    { "type": "integer" },
      "issue_description":{ "type": "semantic_text" },
      "lifetime_value":  { "type": "float" }
    }
  }
}
```

```
{
  "acknowledged": true,
  "shards_acknowledged": true,
  "index": "store-orders"
}
```

store-orders index — 13 customer orders

Shell History Config

Export requests Import requests

```
PUT /return-policies
{
  "mappings": {
    "properties": {
      "category":        { "type": "keyword" },
      "policy_text":     { "type": "semantic_text" },
      "return_window_days": { "type": "integer" },
      "refund_type":     { "type": "keyword" },
      "exceptions":      { "type": "text" }
    }
  }
}
```

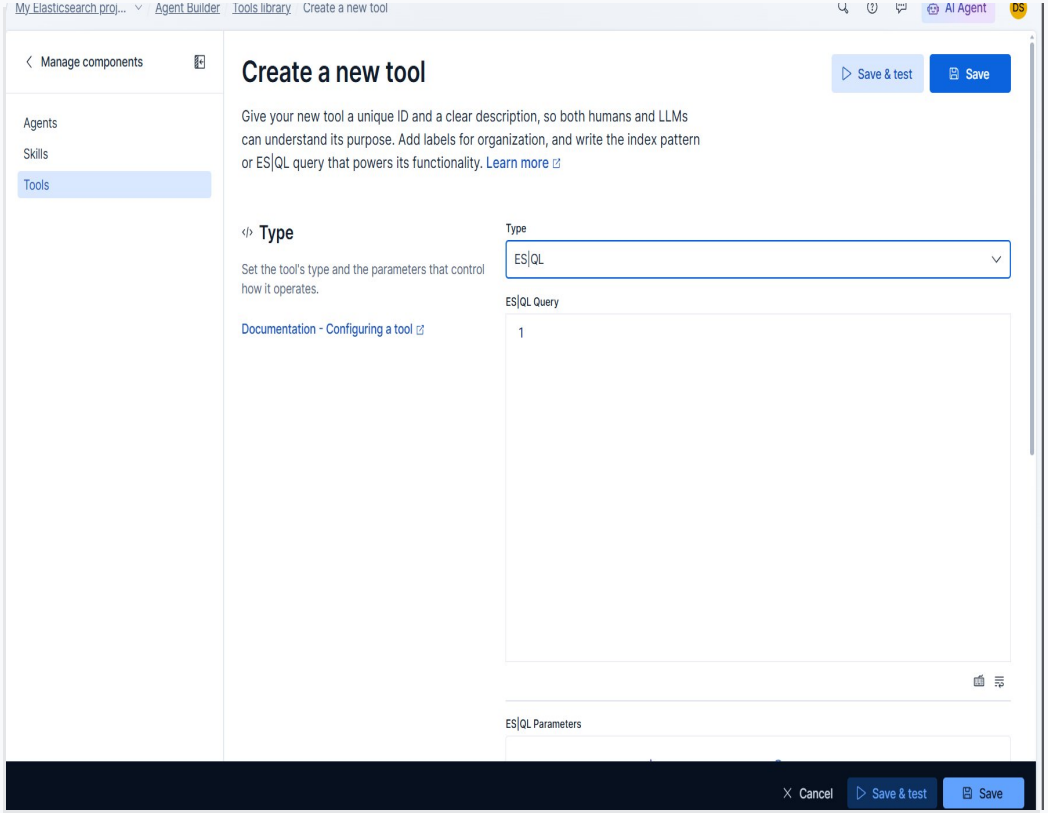
```
{
  "acknowledged": true,
  "shards_acknowledged": true,
  "index": "return-policies"
}
```

return-policies index — 6 policy docs

STEP 3 — TOOLS

Creating ES|QL Tools in Agent Builder

Each tool is defined as an ES|QL query with typed parameters. The LLM reads the description and decides when to call it.



3 tools created:

lookup_order

ES|QL · Typed parameters · Agent-callable

Customer history + lifetime value + return count

get_return_policy

ES|QL · Typed parameters · Agent-callable

Policy rules, refund type, VIP exceptions

find_similar_complaints

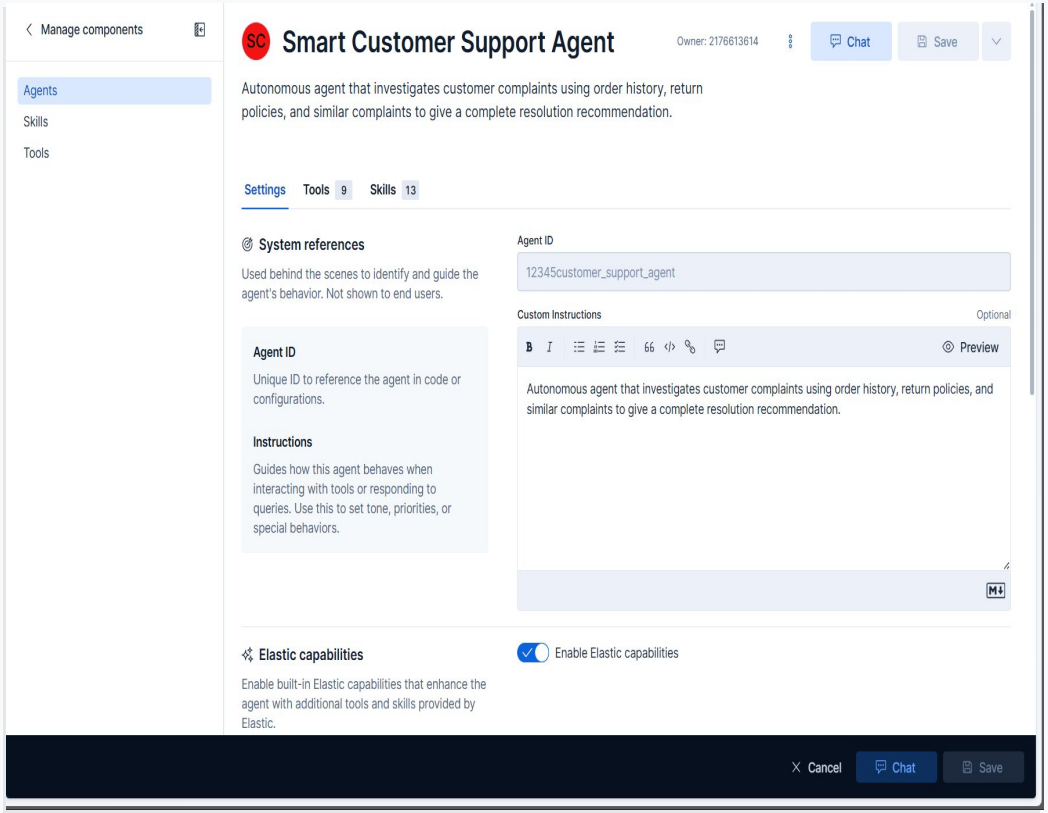
ES|QL · Typed parameters · Agent-callable

Other customers with same issue

STEP 4 — AGENT

Configuring the Smart Support Agent

System prompt defines the investigation process, customer tiers, return abuse rules. Agent follows them autonomously on every query.



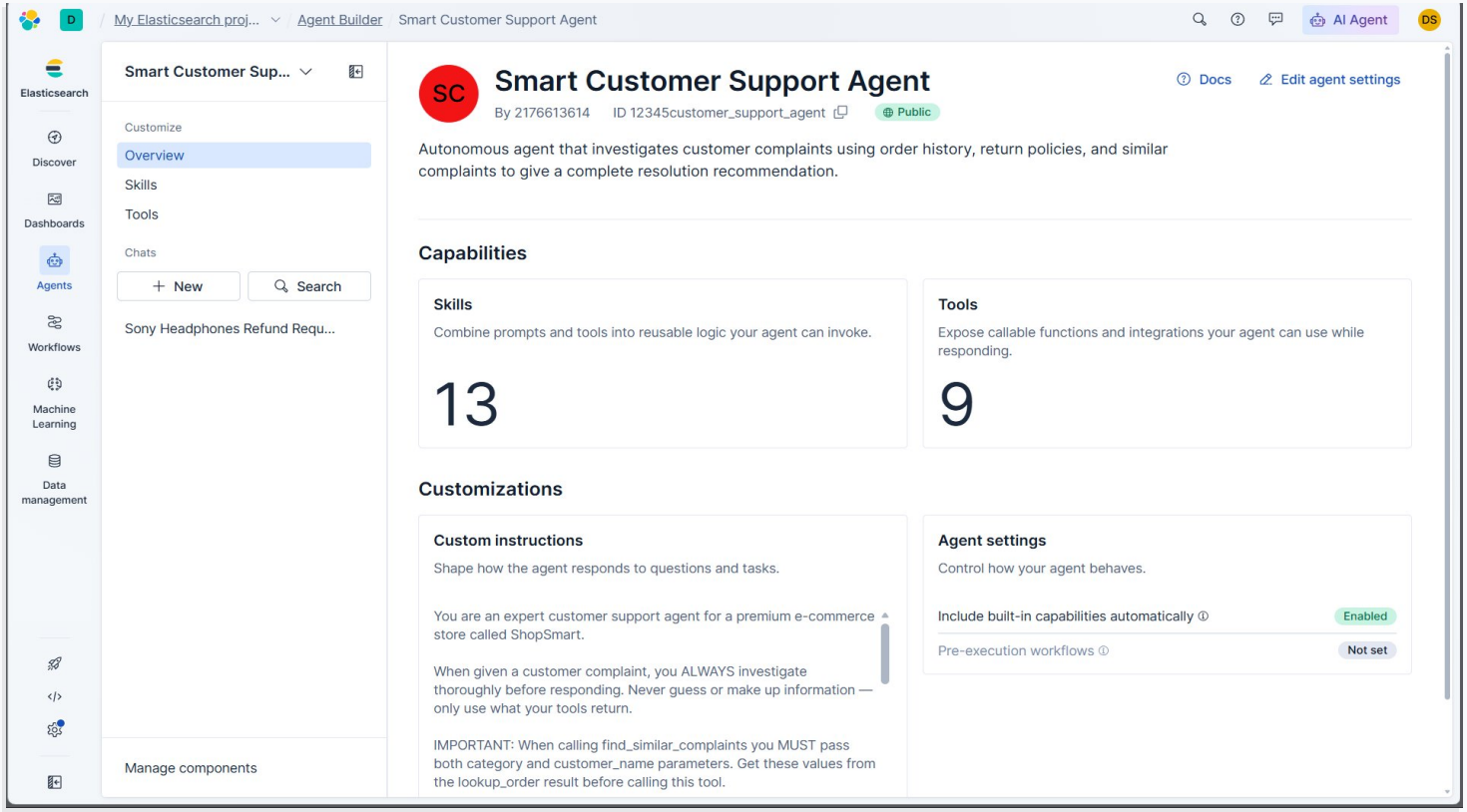
System prompt rules:

- Always call lookup_order first
- Then check return policy
- Then find similar complaints
- VIP if LTV > $2,000
- Flag abuse if returns ≥ 3
- Write the customer email

AGENT LIVE

Smart Customer Support Agent — Live

Agent is Public, has 13 Skills, 9 Tools and a full system prompt. Ready to investigate customer complaints autonomously.



13
Skills

9
Tools

3
Custom tools

0
Lines of code

LIVE DEMO

Smart Customer Support Agent

ShopSmart e-commerce · Elastic Cloud Serverless · Zero lines of code

01

Sarah Johnson

Sony WH-1000XM5 Headphones · $349.99

"Arrived with a cracked headband"

LOYAL · $1,240 LTV

✅ **Full cash refund approved**

0 past returns · damaged shipping · 30-day window

02

Marcus Williams

Nike Air Max 270 · $159.99

"Wrong size delivered, ordered 11 got 9"

STANDARD · $320 LTV

⚠ **Store credit + manager review**

3 past returns = abuse flag · no cash refund

03

Emily Chen

KitchenAid Mixer · $449.99

"Tracking stuck in Chicago 4 days"

VIP · $2,890 LTV

🚀 **Priority reshipment — no return needed**

LTV > $2,000 triggers VIP rule · goodwill gesture

HOW IT WORKS

What the Agent Does — Step by Step

"Sarah Johnson says her Sony headphones arrived with a cracked headband. She wants a refund. What should we do?"

1. **lookup_order called**
   Finds 3 orders for Sarah. Lifetime value: $1,240. Return count: 0 — clean history.
   TOOL CALL

2. **get_return_policy called**
   Fetches Electronics policy: 30-day full refund for damaged items. Also fetches General VIP rules.
   TOOL CALL

3. **find_similar_complaints called**
   Checks other Electronics customers. No similar complaints found — this is an isolated case.
   TOOL CALL

4. **Business rules applied autonomously**
   LTV $1,240 = Loyal tier. 0 returns = no abuse. Within 30-day window. Damaged = eligible for full refund.
   REASONING

5. **Final report + customer email generated**
   Policy ruling, recommended action ($349.99 refund), and ready-to-send email drafted in seconds.
   OUTPUT

CONCEPTS RECAP

What You Just Saw Under the Hood

Inverted index

Words map to documents — search is a lookup not a scan. ES returns results in milliseconds across millions of docs.

Vector embeddings

Text → 384+ numbers. Similar meaning = similar numbers. Finds "broken in transit" when you search "damaged shipping".

Hybrid search + RRF

BM25 keyword + kNN vector run in parallel. Reciprocal Rank Fusion blends scores. Best of both worlds every time.

semantic_text field

Elastic Cloud auto-generates embeddings on ingest. No separate model or pipeline — just declare the field type.

ES|QL tools

Tools defined as typed ES|QL queries. The LLM decides when to call them and what values to pass autonomously.

System prompt = brain

Investigation order, customer tiers, return abuse rules all live in the prompt. Agent follows them every single time.

BUILD TODAY

What You Can Build Today

E-commerce

Fraud Detection Agent

Cross-reference order history, chargebacks, and location. Auto-flag suspicious orders before they ship.

Retail

Customer Support Agent

What you saw today. Orders, policy, history. Correct resolution every time — autonomously.

Analytics

Revenue Drop Investigator

Detect a drop, find which products, correlate with out-of-stock and failed searches. Root cause in seconds.

Operations

Supply Chain Risk Agent

Monitor supplier delays, predict stockouts, prioritise which supplier to call first before Black Friday.

CRM

Churn Prevention Agent

Find VIP customers with exit signals. Cross-reference reviews, tickets, and spend history.

DevOps

IT Ops Intelligence Agent

Detect outages, investigate log cascades, find root cause across microservices. Replaces 3am wake-ups.

Thank You

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3 things to take home:

- LLMs need retrieval to answer domain-specific questions — RAG solves this
- Elasticsearch is the best single tool for hybrid retrieval: keyword + vector + filters
- You can build a working autonomous agent in Kibana today — zero code required

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